

Figure 8.11: Endemic Plants

a. *Bentinckia condapanna* **b.** *Baccaurea courtallensis*

8.8 Carbon Capture and Storage (CCS)

Carbon capture and storage is a technology of capturing carbondioxide and injects it deep into the underground rocks to a depth of 1 km or more and it is an approach to mitigate global warming by capturing CO_2 from large point sources such as industries and power plants and subsequently storing it instead of releasing it into the atmosphere. Various safe sites have been selected for permanent storage in various deep geological formations, liquid storage in the Ocean and solid storage by reduction of CO_2 with metal oxide to produce stable carbonates. It is also known as Geological sequestration which involves injecting CO_2 directly into the underground geological formations (such as declining oil fields, gas fields saline aquifers and unmineable coal have been suggested as storage sites).

Carbon Sink

Any system having the capacity to accumulate more atmospheric carbon during a given time interval than releasing CO_2. Example: forest, soil, ocean are natural sinks. Landfills are artificial sinks.

Carbon Sequestration

Carbon sequestration is the process of capturing and storing CO2 which reduces the amount of CO2 in the atmosphere with a goal of reducing global climate change.

Carbon sequestration occurs naturally by plants and in ocean. Terrestrial sequestration is typically accomplished through forest and soil conservation practices that enhance the storage carbon.

As an example microalgae such as species of *Chlorella*, *Scenedesmus*, *Chroococcus* and *Chlamydomonas* are used globally for CO_2 sequestration. Trees like *Eugenia caryophyllata*, *Tecoma stans*, *Cinnamomum verum* have high capacity and noted to sequester carbon. Macroalgae and marine grasses and mangroves are also have ability to mitigate carbon-di-oxide.

Carbon Foot Print (CFP)

Every human activity leaves a mark just like our footprint. This **Carbon foot print** is the

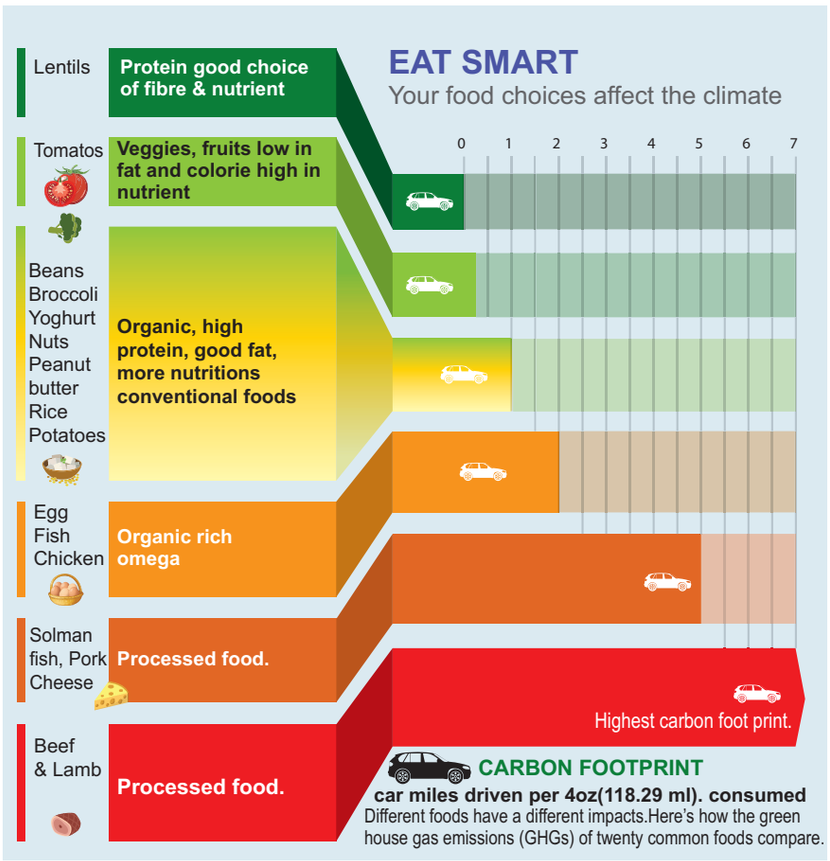


Figure 8.12: Carbon foot print

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